

1(a)	acceleration perpendicular to velocity	B1
1(b)(i)	decreases	B1
1(b)(ii)	(acceleration of) 9.8 m s^{-2} is caused by weight of car or centripetal force must be greater than weight of car	B1
	(acceleration $> 9.8 \text{ m s}^{-2}$) requires contact <u>force</u> from track or (centripetal force $>$ weight) requires contact <u>force</u> from track	B1
1(c)	$\frac{1}{2}mv_Y^2 = \frac{1}{2}mv_X^2 - mgh$	C1
	$a = v^2 / r$	C1
	$v_Y^2 = 3.8^2 - 2 \times 9.81 \times 0.62$ so $v_Y = 1.5 \text{ m s}^{-1}$	A1
	$a = 1.5^2 / 0.31 = 7.3 \text{ m s}^{-2}$ (which is less than 9.8 m s^{-2}) so no	
	or	
	$v_Y = \sqrt{(9.81 \times 0.31)} = 1.74 \text{ m s}^{-1}$ so $v_X^2 = 1.74^2 + 2 \times 9.81 \times 0.62$ $v_X = 3.9 \text{ m s}^{-1}$ (which is greater than 3.8 m s^{-1}) so no	(A1)
1(d)	acceleration is independent of mass so makes no difference or mass cancels in the equation so makes no difference	B1